Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/15/2012
PI Name:	Cooper, Maya M.S.		
Project Title:	Food Processing vs. Packaged Food System Trade Study		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	maya.cooper@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281.483.1892
Organization Name:	Leidos/NASA Johnson Space Center		
PI Address 1:	1300 Hercules MC:C09		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2009	**End Date:**	10/01/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Douglas, Grace	**Contact Phone:**	
Contact Email:	grace.l.douglas@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is 10/1/2012 per M. Perchonok/JSC (Ed., 8/17/2011) NOTE: End date changed to 10/1/2011 for reporting purposes [Ed. 2/28/2011]		
Key Personnel Changes/Previous PI:	Patricia Catauro left the organization, and hence she was not a co-investigator for FY12.		
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

The Food Processing vs. Packaged Food System Trade Study addresses the efficiencies and adequacies of growing produce and processing baseline crops into edible ingredients as compared to the efficiencies and adequacies of the utilizing the existing prepackaged food system. Specifically, the research is parceled into four specific aims: • To determine the requirements on vegetable, fruit, and baseline crops growth and bulk ingredient supply in terms of the estimated pounds required.
• To determine requirements for food preparation and/or processing equipment at the extraterrestrial habitat.

• To provide a quantitative assessment of mass penalties and human factors impacts of each proposed menu option.

Task Description:	• To provide an assessment of feasibility of implementing the proposed systems and identify associated technology gaps. Recent AFT work to prepare for deep space surface missions identified the shortcomings of using the current prepackaged food system in long exploratory mission application. The degradation of the quality of thermostabilized products and degradation of nutritional content indicates the inadequacy of the current food system for a five-year mission. This trade study is central to the decision process on whether to continue work on bioregenerative crop development for future spaceflight missions. Additionally, the trade study supports the decision process on whether to develop food preparation and food processing technologies and, if so, to what extent the development should occur. Relationship to HRP risk(s)/gap(s): This study directly addresses the Advanced Food Technology Risk as given below: Risk of Performance Decrement and Crew Illness Due to an Inadequate Food System: Given there is a constrained spaceflight environment with limited-source food supply, there is a possibility of performance decrement, illness, or loss of mission due to an inadequate food system. Because the food system has huge implications to the upmass considerations of the mission as well as measurable impact to the psyche of the crew members, the appropriate decision on the food system will directly relate to the adequacy of the food system for the mission. At the study conclusion, recommendations will be provided to the Human Spaceflight Architecture Team with regard to the degree of food preparation and food processing that should occur at the Mars surface and other designated habitat locations. A development timeline and preliminary budget for final development will be provided for the recommended food system.
Rationale for HRP Directed Research:	This research is directed because NASA must define complete scientific activities in a short time and there is insufficient time to issue a solicitation.
Research Impact/Earth Benefits:	This project has potential implications in the considerations of the food systems for developing nations as similar tradeoffs between (1) growing and harvesting food and (2) supplying rations of nutritional consequence are made during nation building efforts and disaster relief exercises. This project will quantify on real terms the benefits and risks of both systems. Additionally, the method of determining food system restraint used in this study has wide implications for a common method of assessment of how various food systems (military, disaster, institutional, impoverished areas) might impact human psyche, particularly when combined the standard mood ratings or long-term statistical tracking. To date, only qualitative assessments have been drawn due to lack of formalized rating of food systems. No hardware technology development is expected to result from this project.
Task Progress:	This task was completed in July 2012. The trade study answered the key question to which it was targeted – is it better to take the food to Mars or grow the food on the surface? – with the ambiguous “yes.” While the advantages of growing the food at the surface involve a tremendous upmass savings, higher acceptability, and higher potential menu variety, it cannot be discounted that even a small addition of prepackaged food prompted a significant decrease in crew time utilization and decreased the likelihood of food scarcity. The results indicate the food system design should focus on a solution that is mixed with food sources, leveraging the advantages of both farming crops and stowing the food supply. Five food systems were postulated based on three menu designs of equal caloric contribution. The first food system was based on meals derived from crops grown at the Martian habitat with minor dry ingredient additions. The second food system used the same meals as the first but was based on meals derived from a smaller number of crops grown at the Martian habitat, stable bulk commodities (rice, beans, wheat) shipped from Earth, and minor dry ingredient additions. The third and fourth food systems mirror the first and second system food sources, respectively, except both the third and fourth systems are supplemented with prepackaged meat products. The final food system was based solely on existing space food. Many recipes were reproduced to ascertain the amount of raw material, the necessary cookware, and time that was required to complete a meal. The equivalent system mass (ESM) and the food metric value (FMV) for each of the five food systems were then compared to determine which food system is preferable based upon resource utilization and other food system metrics. For a crew of 6 on a 600-day habitat mission, Food System 3 had the lowest equivalent system mass at 8566 kilograms and the lowest food metric value at 176 kilograms. This food system had both the second lowest mass shipped and second lowest volume shipped but was third in active crew time hours. The ESM for food system 3 was driven primarily by the low mass of stowed ingredients and a crew time utilization that was less than food system 1 or 2. The modified food quality index was not very different between food systems 1-4, and therefore, the value impacted the final results very little. The combination of the food processing data with the ESM calculations of other linked subsystems (e.g. biomass production, water) will present the most accurate view of the food processing cost to mission planners.
Bibliography Type:	Description: (Last Updated: 04/23/2019)
Abstracts for Journals and Proceedings	Catauro PM, Cooper MR. "A Trade Study Comparison of in-Situ-Processed and Pre-Packaged Food Systems." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011
Articles in Peer-reviewed Journals	Cooper MR, Catauro PM, and Perchonok MH. "Development and evaluation of bioregenerative menus for Mars habitat missions." Acta Astronautica. 2012 Dec;81:(2)555–62. http://dx.doi.org/10.1016/j.actaastro.2012.08.035 , Dec-2012

Significant Media Coverage	Rowe A. "Article about Maya Cooper's work at NASA: 'Houston, This Spinach is Delicious: A NASA food engineer prepares the menu for astronauts' first Martian adventure.' " O Magazine, p. 44, Mar 2012., Mar-2012
Significant Media Coverage	Cooper MR, Perchonok MH. "NASA builds Mars menu." Associated Press (AP) news article -- multiple media outlets, July 2012., Jul-2012
Significant Media Coverage	Cooper MR. "NASA Already Planning Meals For 2030 Mars Mission." National Public Radio (NPR) interview of Maya Cooper with Audie Cornish on All Things Considered, July 24, 2012. http://www.npr.org/2012/07/24/157313902/nasa-already-planning-meals-for-2030-mars-mission , Jul-2012
Significant Media Coverage	Witze A. "Space Eats: Astronaut journey to Mars requires new age nibbles." Science News, 2011 Nov 19;180(11):20-1. Article includes interview with Maya Cooper. http://www.sciencenews.org/view/feature/id/335795/title/Space_Eats , Nov-2011